

HEMANT BHANDE

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🐙 GitHub · D LeetCode · _ Codeforces · ' GeeksforGeeks

SUMMARY

Developer and curious problem solver who enjoys understanding how systems work from the inside. Punctual, hardworking, and driven to learn by building. Experienced in backend development, distributed systems, concurrency, and full-stack web applications, supported by a strong foundation in data structures and algorithms.

EDUCATION

Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded	Expected 2027
<i>B.Tech in Information Technology</i>	CGPA: 8.28
Shivaji Mahavidyalaya, Udgir	2020 – 2022
<i>Higher Secondary (PCM), Maharashtra Board</i>	80.67%

TECHNICAL SKILLS

- **Languages:** Java, Go, C++, C
- **Backend & Frameworks:** Spring Boot, Spring Security, Spring Data JPA, Django, REST APIs
- **Databases & Messaging:** PostgreSQL, MySQL, MongoDB, Redis (Streams, Pub/Sub), SQLite
- **Frontend:** React, JavaFX
- **Tools & Concepts:** Docker, Docker Compose, Git, JWT, Multithreading & Concurrency, Distributed Systems, Socket Programming, JUnit, Mockito, Linux/Unix

INTERNSHIP EXPERIENCE

Software Engineering Intern, BodhiTree Platform (Bodhicoder)	May 2026 – Jul 2026
<i>IIT Bombay Trust Lab</i>	

- Built a Bulk CSV Upload and LLM-based edge case generation module for the BodhiTree DSA learning platform using React, Django, and SQLite.
- Designed an LLM-driven pipeline that auto-generates edge cases for DSA problems, reducing manual test-case authoring effort for instructors.
- Implemented bulk test-case upload supporting large volumes of data via ZIP and CSV formats, integrating it into the Django backend and React frontend.

PROJECTS

Distributed In-Memory Key-Value Database (Redis Clone)	🐙 GitHub
<i>Go (Golang), TCP/IP Sockets, RESP Protocol, Radix Tree, Concurrency (Goroutines & Channels)</i>	

- Engineered a Redis-compatible in-memory store in Go, building a multi-client TCP server and a custom byte-level RESP parser to handle pipelined commands and fragmented network buffers.
- Implemented core data structures and TTL management, supporting string operations (SET, GET, INCR with passive expiration), list primitives (LPUSH, RPUSH, LRANGE), and blocking operations (BLPOP) using Go channels.
- Designed a custom Radix Tree (Patricia Trie) from scratch to power Redis Streams (XADD, XRANGE, XREAD BLOCK), supporting auto-generated monotonic IDs and range queries.
- Built advanced Redis subsystems, including atomic transactions with optimistic locking (MULTI/EXEC/WATCH), Pub/Sub message broadcasting, and master-slave replication handshaking.

Distributed Job Scheduling Platform	🐙 GitHub
<i>Spring Boot, PostgreSQL, Redis, React, Docker</i>	

- Architected a distributed job scheduling backend supporting immediate, delayed, scheduled, recurring (cron), and batch jobs, horizontally scaled across multiple Spring Boot instances.
- Implemented atomic job claiming via conditional SQL updates to guarantee exactly-once claiming across concurrent workers, backed by Redis Streams consumer groups.
- Built heartbeat-based lease renewal, recovery services to reclaim stale jobs, and configurable retry policies with dead-letter queue routing.
- Containerized the full stack with Docker Compose and validated behavior with 69 unit, controller-slice, and concurrency integration tests.

Chat Application with Collaborative Whiteboard	🐙 GitHub
<i>Core Java, TCP Sockets, RSA, MySQL</i>	

- Built a custom IoC container and annotation-driven routing framework inspired by Spring to map and invoke controller methods over raw TCP sockets and WebSocket frames.
- Designed a multithreaded TCP server with per-client reader/writer threads for real-time LAN messaging and a collabo-

native multi-room whiteboard with MySQL authentication.

- Implemented RSA key generation, digital signatures, and verification for message integrity.

ACHIEVEMENTS

- Solved 700+ Data Structures & Algorithms and competitive programming problems across LeetCode, GeeksforGeeks, and Codeforces.